

A YOUNG KIM

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EDUCATION

KAIST · M.S. in Data Science · College of Engineering Sep 2024 - Aug 2026
Thesis: Every Answer Has Its Own Path: Agentic Routing for Retrieval-Augmented Table Question Answering (EMNLP 2026 Main)

UC San Diego · B.S. in Probability and Statistics · Department of Mathematics Sep 2016 - Dec 2019

RESEARCH EXPERIENCE

KAIST · Graduate Researcher Sep 2024 - Aug 2026

Agentic RAG & LLM Recommendation · Advisor: Prof. Mun Yong Yi

- Introduced **ATR** (AgenticTableRAG) for hybrid table-text QA, combining sub-query routing and evidence verification to achieve **higher F1 than all evaluated baselines** across three benchmarks under matched retrieval conditions.
- Distilled the routing policy into a 66M-parameter student, matching **98.6% of teacher routing decisions** on HybridQA while eliminating per-sub-query LLM calls for route selection and transferring to unseen benchmarks.
- Developed **R2END** (Reasoning-to-Encoder Distillation) to transfer LLM reasoning into a text encoder through semantic alignment, achieving the **best HR@5 and NDCG@5 among evaluated methods** across four recommendation datasets.

KAIST × Hankook Tire & Technology · Graduate Researcher Sep 2024 - Dec 2025

Multimodal Fusion & Tabular-to-Image

- Implemented and trained **TAMIG** (Tabular-Aware Multimodal Image Generation), a two-stage PyTorch framework fusing tire specifications, 1D mold profiles, and 2D pattern images; refined outputs with JAX-based Thera inference.
- Accelerated the overall tire manufacturing pipeline by replacing physical testing with AI-generated prototypes, contributing to **significant reductions in production time and operational costs**.

Korea University · Research Intern Feb 2024 - Jul 2024

Novel View Synthesis & 3D Generation · Advisor: Prof. Sangpil Kim

- Investigated **NeRF and 3D Gaussian Splatting** for novel view synthesis and 3D scene representation, implementing and evaluating AI watermarking methods to verify ownership of reconstructed scenes.

WORK EXPERIENCE

KPMG Korea · Digital Consultant · Seoul, South Korea Dec 2021 - Nov 2023

- Authored and presented proposals directly to **senior client decision-makers**, including department directors.
- Delivered data preparation, ML model development and deployment, and digital strategy execution across banking and retail engagements.

NongHyup, Big Data All-In-One Platform N-Hub Implementation Mar 2023 - Nov 2023

- Managed the end-to-end product planning of a HanaroMart analytics service for store managers, taking full charge of client meetings, service design, visualization engineer management, and final rollout presentations.
- Built data mart using Apache Hadoop, and engineered TensorFlow models for agricultural **price forecasting**.

Major Bank, Data Analysis System Implementation Dec 2022 - Feb 2023

- Architected a scalable **SQL data mart** to consolidate Special Account Surplus Fund, facilitating data-driven executive decision-making and operational monitoring through the deployment of customized dashboards.

Major Bank, AI Recommender System Implementation Using MyData Aug 2022 - Dec 2022

- Developed transaction-based customer **clustering and recommendations** for wealth management and banking products, contributing to a major bank's MyData service reaching **3.55M+ users**.

NongHyup, NH Big Data Platform Implementation Dec 2021 - May 2022

- Built **prediction models** for smart-farm simulation systems; clustered and visualized HanaroMart store types and analyzed product affinities to profile stores and their sales patterns.

LG Electronics North America · Software Engineer · San Diego, United States

Mar 2020 - Nov 2020

- Built Python regression suites covering **100+** network, connectivity, and battery scenarios across LG mobile-device configurations, expanding automated release validation and reducing manual testing.
- Authored model-specific test plans, reproduced defects, and documented traceable evidence for engineering teams to diagnose and resolve regressions before release.

ENTREPRENEURIAL EXPERIENCE

Finnote · Founder & AI Engineer · finnote.io

Mar 2026 - Present

- Develop **Finnote**, an investment research service that transforms DART and SEC filings into verifiable answers linked to exact source passages, applying **ATR's query routing and evidence verification methods**.
- Lead product direction and the full filing pipeline from ingestion and extraction through retrieval, citation tracking, and interface design, and build workflows that turn filings into investor reports and Instagram visual briefings.

Rimbic, Inc. · Founder & AI Engineer · [Code](#) · [Demo](#)

Jun 2024 - Jun 2025

- Built **Rume**, a standalone AI smart speaker that proactively engages users, combining a low-latency voice agent (OpenAI Realtime API, Agora) with a fine-tuned Stable Video Diffusion model deployed on Jetson Nano and Raspberry Pi.
- Secured **KRW 47M (~USD 35K)** in non-dilutive funding through the government's **Pre-Entrepreneurship Package** and led product development, hardware-software integration, and go-to-market execution.

PUBLICATIONS

Every Answer Has Its Own Path: Agentic Routing for Retrieval-Augmented Table Question Answering

Accepted to EMNLP 2026 Main Conference · [Code](#) · [Paper](#)

[A Young Kim](#), Jisu Shin, Donghee Han, Mun Yong Yi

SAFE: A Plug-and-Play Framework for Prompt Optimization in Text-to-Image Safety

Accepted to EMNLP 2026 Main Conference

Jisu Shin, Sol Lee, Sungrae Hong, [A Young Kim](#), Mun Yong Yi

Distilling LLM Reasoning into Dense Encoders: Bridging the Accuracy-Efficiency Gap in Recommendation

ACL 2026 Findings · [Paper](#)

Donghee Han, Daeyoung Roh, [A Young Kim](#), Hwanjun Song, Mun Yong Yi

DRCoD: Toward Robust Continual Learning of Diffusion Models for Tire Manufacturing Prototyping

CVPR 2026 Workshop · [Paper](#)

Jisu Shin, Sol Lee, Sungrae Hong, [A Young Kim](#), Younbin You, Jeongheon Park, Jungsoo Oh, Mun Yong Yi

SKILLS

Programming: Python, Java, R, Bash/Zsh, MATLAB

ML & Serving: PyTorch, JAX, TensorFlow, Keras, Hugging Face, Vertex AI, vLLM

LLM Systems: Agentic RAG, Knowledge Distillation, Table QA, Text-to-SQL, Diffusion models, Multimodal Fusion

Data & Analytics: SQL, Apache Hadoop (Hive, Impala), Oracle, MySQL, Tableau, SPSS

Languages: Korean (native), English (full professional proficiency)

CERTIFICATES

Korea Data Agency (K-DATA): Big Data Analysis Engineer, Data Architecture Semi-Professional (DAsP), SQL Developer (SQLD), Advanced Data Analytics Semi-Professional (ADsP)

Statistics Korea: Survey Analyst